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RESEARCH ARTICLE

Acute and chronic sublethal chemical pollution affects activity, learning and memory in mosquito larvae

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ABSTRACT

Freshwater ecosystems play a critical role in supporting biodiversity and providing essential environmental services. However, these ecosystems are increasingly threatened by human activities, including habitat loss, pollution and climate change. Traditional assessment methods focus on water properties, but biomonitoring approaches, particularly those examining behaviour and cognition, provide valuable insights into the ecological effects of pollutants. This study examines the effects of three common pollutants (glyphosate, atrazine and paracetamol) on the cognitive abilities of *Aedes aegypti* mosquito larvae, a vector for several diseases. We used an automated bioassay to study habituation learning and the effects of the three pollutants alone or in combination, at sub-lethal doses ranging from field-realistic to commercially recommended levels. Our results show that the three compounds modulate individual spontaneous activity and impair habituation and memory retention. These changes may alter the perception or the behavioural response of mosquito larvae to signals of their environment as indicating the presence of conspecifics or predators, and suggest that other organisms living in freshwater ecosystems may also be affected. Incorporating behavioural and cognitive assessments in ecotoxicological studies provides a more comprehensive understanding of the ecological effects of pollutants, which is needed to address economic challenges in fragile ecosystems.

KEY WORDS: Habituation, *Aedes aegypti*, Paracetamol, Acetaminophen, Glyphosate, Atrazine, Cocktail effect

INTRODUCTION

Freshwater ecosystems provide essential services, support 10% of the world's animal species and represent a critical reservoir of biodiversity (Madhav et al., 2020; Zhang et al., 2021). Yet, they are among the most endangered environments worldwide, threatened by habitat degradation, overexploitation, invasive species, climate change, hydrological alterations and widespread use of chemical compounds for industrial, agricultural and domestic purposes (Sala et al., 2000; Arthington, 2021; Vári et al., 2022). Traditional monitoring relies on sampling water and sediment and measuring their physical and chemical properties (Bartram and Ballance, 1996; Zhou et al., 2008), but these approaches often fail to capture interactions among pollutants or subtle effects on the biodiversity living in these ecosystems (López-López and Sedeño-Díaz, 2015; Zaghloul et al., 2020; Malafaia and Da Luz, 2024).

Behavioural and cognitive biomonitoring is emerging as a complementary approach to link physiological processes with ecologically relevant outcomes (Blaxter and Hallers-Tjabbes, 1992; Melvin and Wilson, 2013; Bownik and Wlodkowic, 2021). Assessing behavioural traits in an eco-ethological context allows evaluation of pollutant impacts on ecological drivers of animal adaptation and provides early indicators of sublethal stress (Hellou, 2011; Hong and Zha, 2019). Despite these advantages, standardised, inexpensive, reproducible and non-invasive methods for behavioural and cognitive bioassays are still lacking (Jacquin et al., 2020; Ford et al., 2021; Bertram et al., 2022).

Aedes aegypti is a globally distributed vector of numerous human and animal diseases (i.e. dengue, chikungunya, Zika, yellow fever, West Nile virus, Dirofilariasis) and a widely used laboratory model for behavioural and toxicological studies (Kraemer et al., 2019; Adegoke et al., 2020). Adults have well-characterised resistance mechanisms to insecticides (Rubio-Palis et al., 2023). For instance, exposure to atrazine or glyphosate induces the expression of cytochrome P450 monooxygenases (P450s), glutathione *S*-transferases (GSTs) and carboxylesterases, which can enhance larval tolerance to insecticides (Bara et al., 2014).

By contrast, larval stages are less studied but could represent excellent models to investigate pollutant effects on behaviour and cognition. They are easy to rear, inexpensive to maintain and develop rapidly within 4–10 days. Their behaviour is well characterised (e.g. locomotion, feeding, anti-predation responses), making them suitable for reproducible bioassays. They display learning and memory abilities that are likely to be sensitive to pollutants at sublethal levels (Boyer et al., 2006; Baglan et al., 2018; Black et al., 2021). Larvae develop in small water-filled containers, which are often contaminated by agricultural runoff, where herbicides and agrochemicals are transported during rainfall events, or aerial deposition, where chemicals spread via aerosols or dust and settle onto breeding sites (Lima et al., 2022; Upshur et al., 2023).

Among globally widespread pollutants, glyphosate and atrazine are heavily used herbicides that persist in aquatic ecosystems and have documented neurotoxic effects (Battaglin et al., 2014; Li et al., 2018), resulting in an alteration of locomotion, learning and memory in mammals, amphibians and teleost fishes (Rohr and McCoy, 2010; Gill et al., 2018; Rastegar-Moghaddam et al., 2019; Abdulelah et al., 2020), nematods (García-Espiñeira, 2018) and arthropods (Bara et al., 2014; Baglan et al., 2018; Bataillard et al., 2020; Araújo et al., 2021; Kanabar et al., 2021; Weidenmüller et al., 2022; Nouvian et al., 2023). Paracetamol (acetaminophen), the most widely used drug worldwide, is frequently detected in the environment as a result of excretion in urine and release into wastewater (Duong et al., 2016; Hider–Mlynarz et al., 2018; McCrae et al., 2018). It can accumulate in small water-filled containers via greywater and poor wastewater, and medical waste

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treatment (Gracia-Lor et al., 2012). Although few behavioural assays exist, a study in the zebrafish *Danio rerio* showed toxic effects of paracetamol on individual morphology, pigmentation, locomotion, enzyme expression and epigenetics (Nogueira et al., 2019). Altogether, these three chemicals are widespread and persistent, and despite clear evidence on their neurotoxic properties on vertebrates, their impact on behaviour and cognition in larval insects remains poorly understood (Hodkinson and Jackson, 2005; Bonada et al., 2006; Li et al., 2019; Previšić et al., 2020; Malhotra et al., 2021).

In our laboratory, we established a procedure for evaluating cognition in mosquito larvae several years ago (Baglan et al., 2017) that allowed us to characterise the impact of pollutants on learning and memory (Baglan et al., 2018). The system was later automated to eliminate observational bias and obtain precise quantitative data (Dessart et al., 2023). This system was shown to be a powerful tool for studying fundamental cognitive processes (Dessart et al., 2024), as well as the impact of pollutants on mosquito behaviour (this work).

In this study, we assessed the sublethal effects of these three compounds, individually and in combination, on locomotion, learning and memory in *Aedes aegypti* larvae using a standardised behavioural assay (Baglan et al., 2017; Dessart et al., 2023, 2024). By looking at acute and chronic exposure, we captured ecologically relevant impacts of pollutants that underlie larval behaviour and cognition as a potential biological indicator. Our fine-scale quantitative analysis provides an innovative, cost-effective and scalable tool for behavioural ecotoxicology.

MATERIALS AND METHODS

Animals

Mosquito eggs (*Aedes aegypti* Bora strain) were obtained from MIVEGEC-IRD (Montpellier, France). Eggs were kept dry or placed in 750 ml polypropylene containers with either 500 ml dechlorinated tap water or 500 ml mixture (see 'Chemical treatment', below). Larvae were fed *ad libitum* with shrimp food (JBL Novo Prawn, Neuhofen, Germany) and the containers were maintained at 25±2°C, under a 12 h:12 h light:dark cycle (lights on at 08:00 h). Only fourth instar mosquito larvae were trained in the experiments. All animals were reared and manipulated in accordance with European Union ethical guidelines.

Experimental procedure

All experiments were performed in the same experimental room at 25±0.5°C during the afternoon (i.e. from 12.00 to 19.00 h). The experimental apparatus was the same as that used in two previous studies and consisted of a platform for isolating individual mosquito larvae, different types of stimuli and a system for recording and analysing larval behaviour (Dessart et al., 2023, 2024). It included individual containers (i.e. 1.5 ml spectrophotometer plastic cuvettes) filled with treated water (see 'Acute toxicity', below) or dechlorinated tap water (Fig. 1A). Fourth instar larvae were placed in 10 horizontally aligned cuvettes and exposed to an aversive stimulus consisting of either a black cardboard square (16 cm side) attached to a servomotor, or four mechanical vibrators, all operated

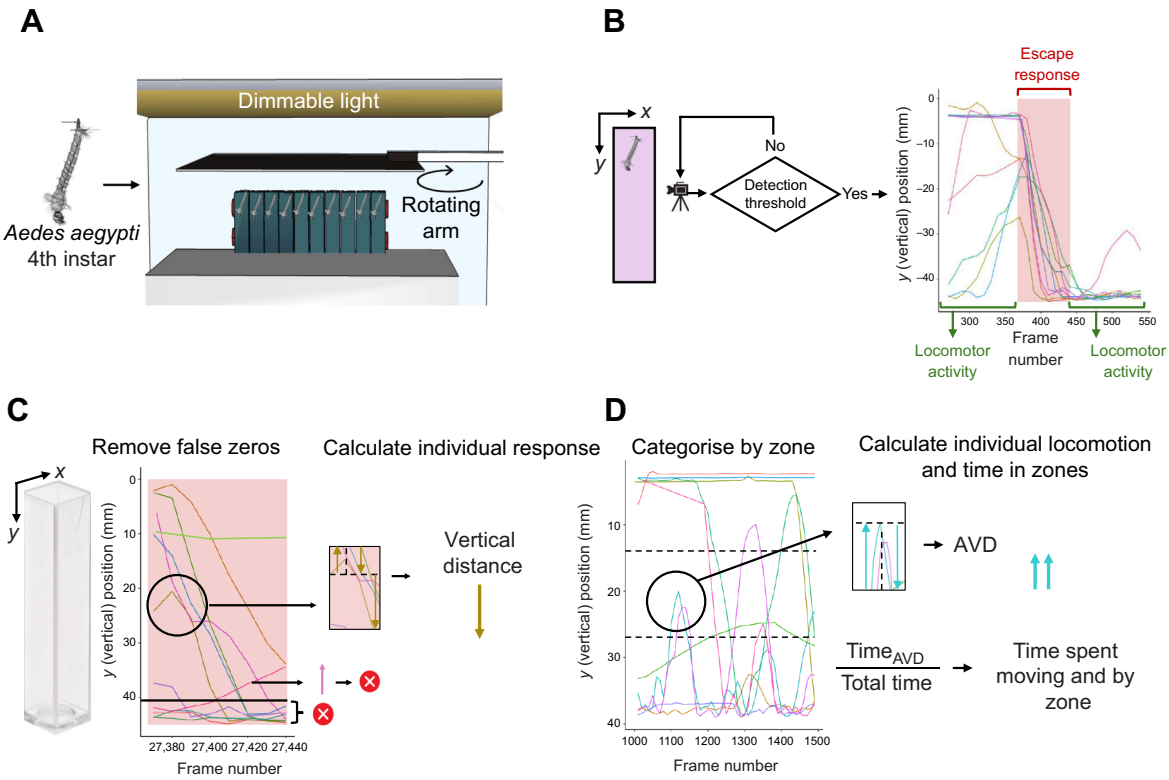


Fig. 1. Schematic diagram of the experimental protocol. (A) *Aedes aegypti* larvae were reared in the laboratory and trained at the 4th larval stage using our training apparatus. (B) The acquisition, filtering, extraction and classification process. Experiments were video-recorded and individual trajectory was extracted. (C) Escape response. We analysed the stimulation response that corresponded to the individual trajectory during the aversive stimulus, using the vertical distance. This quantitative variable was calculated as the relative sum of the distance travelled in the vertical direction toward the bottom of the cuvette. In addition, two filters were applied to exclude individuals located at the bottom of the cuvette during the first frames of the aversive stimulus and the individuals travelling upwards during the stimulus. (D) Locomotor activity. We analysed the spontaneous locomotor activity using three variables: time spent per zone, the proportion of time spent in one of the three delimited zones; time spent moving, the proportion of time where the vertical distance (VD) travelled was above a threshold of 1 mm s^{-1}; and absolute vertical distance (AVD; quantitative), calculated as the absolute sum of the distance travelled in the vertical direction.

by an Arduino Uno board remotely controlled by a computer. The cardboard represented a potential flying predator visually perceived by the larvae, while the vibrator represented the motion of potential fish predators transmitted through the water column. The cuvettes were illuminated by two LED panels and the light intensity was fixed at 900±100 µW cm^{-2} (International Light Technology radiometer). All experiments were video-recorded, and the videos were subsequently analysed (Fig. 1B, see 'Data analysis', below).

After 30 min of familiarisation, 10 trials were performed, i.e. 10 passages of the visual stimulation, with a 2 min inter-trial interval. For the acute toxicity (see below), a new trial was performed 2 h after the 10th trial to test the memory persistence of mosquito larvae. After training, we applied a 4 s mechanical stimulation using the vibrators to check that the larvae were still responsive to a stimulus. Individuals that did not respond to the mechanical stimulus were excluded from the analysis (7 individuals). We also removed individuals that remained motionless during the training (10 individuals), transformed into pupae during the training (1 individual) and for which, when tracking, we failed to extract individual coordinates (1 individual). A total of 765 individuals were kept for the analyses.

Chemical treatment

Two types of toxicity were represented in this study. Chronic toxicity represents the effect of exposure at low concentrations over the development from egg to 4th instar, whereas acute toxicity was achieved by a single exposure to higher concentrations of chemicals (Environmental Protection Agency, 1994). We simulated chronic toxicity by placing eggs in 750 ml polypropylene containers with either 500 ml of water purified using a Millipore Milli-Q lab water system or 500 ml of treated water, until the fourth larval stage. To represent acute toxicity, larvae were reared in 500 ml purified water, and they were subsequently exposed to treated water or purified water (during 30 min familiarisation+training period). Three chemicals were considered for both toxicity simulations: glyphosate, atrazine and paracetamol. Glyphosate solutions were prepared by dissolving glyphosate (PESTANAL analytical standard, purity ≥98%, Sigma-Aldrich), atrazine (PESTANAL analytical standard, purity ≥98%, Sigma-Aldrich) and paracetamol (BioXtra, purity ≥98%, Sigma-Aldrich) in Millipore Milli-Q lab water. We used the pure chemical rather than a formulation to isolate any possible effect due to these substances being in a mixture. Indeed, formulations with several compounds have been shown to be more toxic than the chemical alone (Nagy et al., 2020).

Chronic toxicity

To investigate the chronic toxicity of the two herbicides (glyphosate and atrazine) and the medicine drug (paracetamol), mosquito larvae were reared in treated water. For glyphosate, we selected two concentrations related to field measurements: 100 and 200 µg l^{-1} (Struger et al., 2008; Byer et al., 2008; Riaz et al., 2009) and a high concentration was prepared at 2 mg l^{-1}, which corresponds to usual doses applicated in other work (Bara et al., 2014; Balbuena et al., 2015; Baglan et al., 2018; Nouvian et al., 2023). For atrazine, three concentrations were selected: 200 µg l^{-1} and 500 µg l^{-1} and 2 mg l^{-1}, consistent with several publications (Dewey, 1986; Bara et al., 2014; Johnson et al., 2019; Abdulelah et al., 2020; Adedara et al., 2021). For paracetamol, although field concentrations reported by some studies were relatively low (e.g. Gracia-Lor et al., 2012: median 44.8 µg l^{-1}, minimum–maximum values 1.13–201 µg l^{-1}), we chose to test concentrations of 1, 10 and 100 mg l^{-1} in line with previous bioassays in mammals (Adamson et al., 1991; El Menyiy et al., 2018), in the zebrafish *Danio rerio* (Nogueira et al., 2019) and on the development of the fly *Calliphora vicina* (O'Brien and Turner, 2004).

A control treatment using only purified water (hereafter referred to as 'water treatment') was paired with each set of experimental conditions. A first water treatment (no. 1) was associated with 200 and 500 µg l^{-1} atrazine, and a second water treatment (no. 2) was associated with 100 and 200 µg l^{-1} glyphosate, and with 1 and 10 mg l^{-1} paracetamol. This first experiment was called 'field doses' because it represented low concentrations associated with field measurements. A new water treatment (no. 3) was associated with 2 mg l^{-1} atrazine and another one (no. 4) with 2 mg l^{-1} glyphosate and 100 mg l^{-1} paracetamol. This second experiment was called 'spray doses' because it represented high concentrations associated with field applications.

In a second set of experiments, we compared the effect of chronic toxicity of chemicals alone or in mixtures alongside the water treatment. The first mixture consisted of concentrations measured directly in water, in the region Centre-Val de Loire, France. A paracetamol concentration of 6 µg l^{-1} corresponded to the maximum measured in the Loire River basin between 2018 and 2019 (Ledieu et al., 2021); the concentrations of 0.126 µg l^{-1} atrazine and 2.3 µg l^{-1} glyphosate corresponded to the highest concentrations found at monitored sites in the Indre et Loire area between 2019 and 2021 (https://naiades.eaufrance.fr/). These concentrations were communicated by the Institute of Organic and Analytical Chemistry (ICOA UMR 7311 CNRS – Université d'Orléans).

The second mixture consisted of 200 µg l^{-1} atrazine and 1 mg l^{-1} paracetamol, two treatments that did not affect individual learning (see Results). A water treatment (no. 5) was associated with the individual chemicals, and an additional water treatment (no. 6) was associated with the two mixtures, in this set of experiments called 'field-realistic doses' *sensu* Herbert et al. (2014).

Acute toxicity

To assess acute toxicity, larvae were reared in purified water and exposed to treated water only during training, to simulate a short-term exposure scenario at high concentration (e.g. representing an aerial spraying). We used the same concentrations as the spray doses used in the chronic toxicity tests (glyphosate and atrazine: 2 mg l^{-1}, paracetamol: 100 mg l^{-1}) to allow direct comparison and ensure detectable behavioural effects within a short time frame. We analysed the effects of these treatments on individual learning, and we also investigated whether these treatments would affect individual memory retention by applying a new trial 2 h after the training session. This retention time was chosen based on our previous work showing that larvae could retain information for up to 2 h after visual learning (Dessart et al., 2023). A last water treatment (no. 7) was associated with these experiments.

Data analysis

For each experiment of 10 individuals, two sets of videos were recorded. First, we recorded the 30 min familiarisation period and compared global spontaneous activity between groups by taking the last 5 min of this recording. Then, we recorded the whole duration of the training period. The videos were analysed similar to those in Dessart et al. (2024). Briefly, each individual trajectory was extracted using tracking software and larval identity and detection rate were verified (Fig. 1B; Tables S1 and S2). Individual vertical positions were smoothed and collapsed to reduce the data number, classified into two categories and analysed separately (Fig. 1). First, the escape response lasted for the 3 s period during which the visual

stimulus appeared above the individuals (Fig. 1C). These data were filtered within each trial response to remove false zeros (i.e. when the larvae were already at the bottom of the cuvette at the start of a trial, see Dessart et al., 2024; Table S3). After this step, we defined the vertical distance as the response variable corresponding to the escape response. This variable was 0 if the individual did not move during a stimulus period, and increased as the individual moved away from the stimulus (i.e. dived downwards; Fig. 1C). Second, locomotor activity represented individual spontaneous activity during the familiarisation and the inter-trial interval (ITI) period (Fig. 1D). These data were outside the stimulus period and were therefore not filtered. We analysed locomotor activity using the absolute vertical distance, which measured individual displacement irrespective of the direction of movement (upward or downward). This variable was more consistent in representing the average speed and number of diving events of an individual across the ITIs. Finally, we also divided the cuvette in three equal zones (Fig. 1D) and calculated the time spent per zone and the time spent moving. To ensure that individual size did not influence locomotor or learning performance, we assessed the relationship between the average silhouette area derived from the tracking method and both the mean locomotor speed (Fig. S16) and the latency to respond in the first trial (Fig. S17), and found no significant correlations.

Statistical analyses

Learning performance

We modelled the learning performance using the generalised additive model to provide a visual estimate of the behavioural response across trials during the training period. We defined models of increasing complexity and compared the best smoothing function using the GCV-UBRE criterion from the *mgcv* package (Wood, 2017). Similar to our previous work, the best smoothing function was the P-spline and was applied to all smoothing curves (Dessart et al., 2024).

For each treatment, we compared the distance travelled during a stimulation in the first and last trials. If the difference was not statistically significant, we concluded that the individuals had not habituated. To make these comparisons, we applied a linear mixed effects model to compare the response at the 1st trial versus the 10th trial. The water treatments were analysed similarly to verify that the individuals reared and trained in clear water were able to habituate. We chose vertical distance as the response variable, trial as a fixed factor and individual identity as a random factor. We checked variance homogeneity and the distribution of residuals using the *simulateResiduals* function from the *DHARma* package (https://CRAN.R-project.org/package=DHARMa). For the assessment of acute toxicity, we applied the same model but including the response at the test phase. For these models, we evaluated pairwise comparisons using the *emmeans* package with Tukey correction (https://CRAN.R-project.org/package=emmeans).

Spontaneous activity

We compared individual spontaneous activity during familiarisation, considering the last 5 min of each treatment. We also compared the nine ITIs between the treatments. First, the absolute distance travelled by individuals was averaged to calculate the individual average speed (mm s^{-1}) (Fig. 1D). We divided the cuvette into three equal zones (top, middle, bottom; Fig. 1D), and counted the time spent in each zone (%). We also confronted absolute vertical distance with a threshold of 1 mm s^{-1} and classified our data into the variable time spent moving (Fig. 1D). Furthermore, we counted the number of diving events by creating a function that counted each time an individual crossed 1/3 and 2/3 of the cuvette length (i.e. 120 pixels) on its way in and out (Fig. 1D). For the average speed, the time spent moving and the number of diving events comparison, we applied a one-way ANOVA with, respectively absolute vertical distance, time spent moving or diving events as the response variable and treatment as factor, followed by Tukey's *post hoc* test for multiple comparisons.

RESULTS

For each treatment, we assessed (i) learning, by comparing the response during the 1st trial (initial stimulus encounter) with that during the 10th trial, and (ii) spontaneous activity, by comparing average speed, time spent moving and number of diving events between treatments during the familiarisation phase (i.e. before training) and the training period (i.e. the 2 min inter-trial intervals). A full summary of learning results for each experiment is provided in Table S4, and all spontaneous activity results are detailed in Table S5.

Chronic exposure to field doses

To simulate chronic exposure, mosquito larvae were reared in water containing different concentrations of glyphosate, atrazine or paracetamol. We first compared low and high concentrations corresponding to field-relevant doses reported in the literature (see Materials and Methods). At a low dose of glyphosate and a high dose for all three chemicals, no significant difference was observed between the 1st and the 10th trial, suggesting that these pollutants may impair an individual's ability to modulate the behavioural response over time ($P>0.05$; Fig. 2; Table S4). Larvae reared in atrazine also moved faster and dived more frequently than those in water treatment no. 1 during both the familiarisation and training periods ($P<0.05$; Fig. 3; Table S5, Figs S1–S4, S9–S12). Larvae were more prone to dive and did so more rapidly, resulting in increased energy expenditure.

Chronic exposure to spray doses

To reflect exposure to agrochemical sprayed doses, we then reared larvae in a third, higher concentration of each chemical alone. Atrazine and glyphosate at spray doses confirmed their negative impact on learning, with prolonged and stable responses observed throughout training ($P>0.05$; Fig. 4; Table S4). These deleterious effects were associated with increased locomotion activity for glyphosate, reflected in higher diving frequency and locomotion speed ($P<0.001$; Fig. S5 and Table S5). Together, these two experiments are consistent with our previous findings on the effect of glyphosate on learning, validating our methodological approach and providing a robust dataset to distinguish neurotoxic effects affecting learning from broader alterations in general activity (Baglan et al., 2018).

Chronic exposure to field-realistic doses

Freshwater pollutants are often present as mixtures, potentially producing additive or synergistic effects (Gilliom, 2006). After identifying field-relevant doses, we selected two concentrations (atrazine at 200 µg l^{-1} and paracetamol at 1 mg l^{-1}) that, alone, revealed no effect on learning (see 'Chronic exposure to field doses', above). When combining these two doses, the mixture ('mixture 1') produced a marginal effect on learning ($P=0.0518$; Fig. 5; Table S4) and consistently increased diving frequency and locomotion speed, similar to the effects observed with atrazine alone at 200 µg l^{-1} ($P<0.05$; Table S5, Figs S7 and S14).

With support from the Institute of Organic and Analytical Chemistry (UMR 7311, Orléans, France), we designed a second mixture to simulate chronic exposure at field-realistic levels. Concentrations of the three chemicals were measured directly in water, and we analysed the effects of single chemicals versus the three chemicals combined. While

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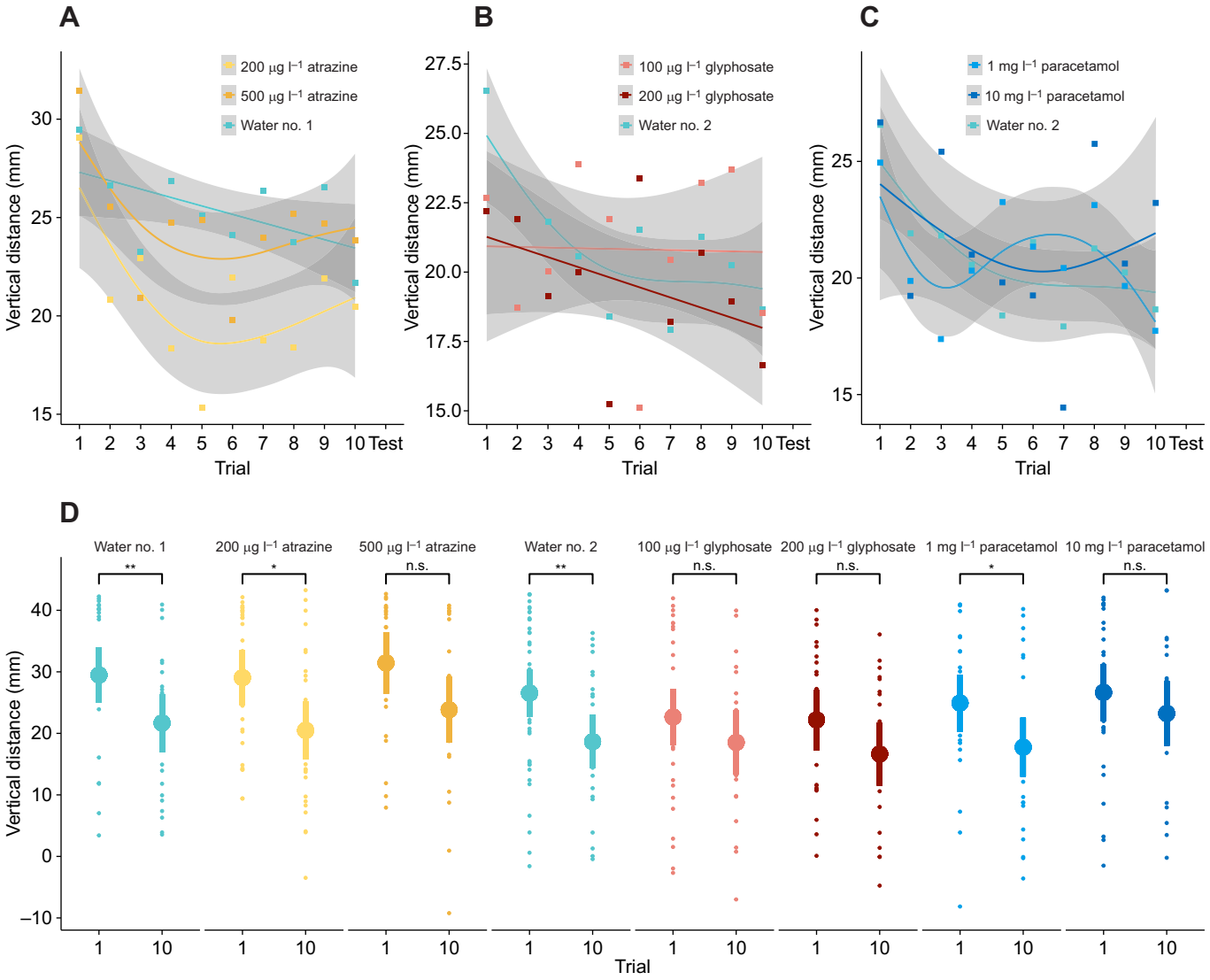


Fig. 2. Learning performance for larvae reared at field doses. (A–C) Habituation curves for larvae reared in water treatment (cyan; for definition of treatment numbers, see Materials and Methods), 200 or 500 µg l^{-1} atrazine (yellow; A), 100 or 200 µg l^{-1} glyphosate (red; B) and 1 or 10 mg l^{-1} paracetamol (blue; C). Vertical distance corresponds to the distance travelled by one individual during the stimulus period (3 s), from the 1st to the 10th trial. Smoothing lines indicate the best fitted GAM model. Squares indicate mean values and grey shading indicates the 95% confidence interval for the average response. (D) For each treatment, vertical distance travelled by individuals responding to an aversive stimulus during the 1st to the 10th trial. Large circles indicate mean values and bars indicate ±95% confidence intervals. n.s., not significant; *P<0.05, **P<0.01.

larvae exposed to individual pollutants were able to habituate, the mixture of these three pollutants at the same concentrations disrupted learning (P=0.35; Fig. 5; Table S4) without affecting locomotion (Table S5, Figs S6, S7, S13 and S14). These experiments provided key insights about the combinatorial neurotoxic effects of these widespread chemicals under realistic exposure scenarios.

Acute toxicity

The presence of chemicals can be related to chronic and long-term exposure but also to high concentration and short-term and acute exposure, for instance during aerial or ground spray, runoffs after rainfall or wastewater surges (Madhav et al., 2020). To simulate such strong acute exposure, we reared larvae in purified water and exposed them to pollutants only during the training period, at spray doses used in the chronic exposure experiments. We evaluated the learning performance and, considering that larval memory retention lasts between 2 and 3 h, we performed an additional trial 2 h after the training period. By comparing the response at the 1st trial with that at this additional test phase, we obtained an estimation of another essential cognitive function: remembering that a stimulus becomes irrelevant, thereby conserving energy (Dessart et al., 2024). No significant effect on learning was found during this short exposure (P<0.03; Fig. 6; Table S4). However, larvae exposed to glyphosate and atrazine showed increased responses in the additional test phase, suggesting a lack of memory retention (P=0.25 and P=0.815; Fig. 6; Table S4). No major changes in spontaneous activity were detected except a slight decrease in time spent moving for larvae exposed to paracetamol (Table S5, Figs S8 and S15). Overall, this final set of experiments highlights the deleterious effects of brief pollutant exposure on larval memory abilities.

DISCUSSION

The development of behavioural tests to assess the effects of pollutants at field-realistic concentrations on the cognitive abilities

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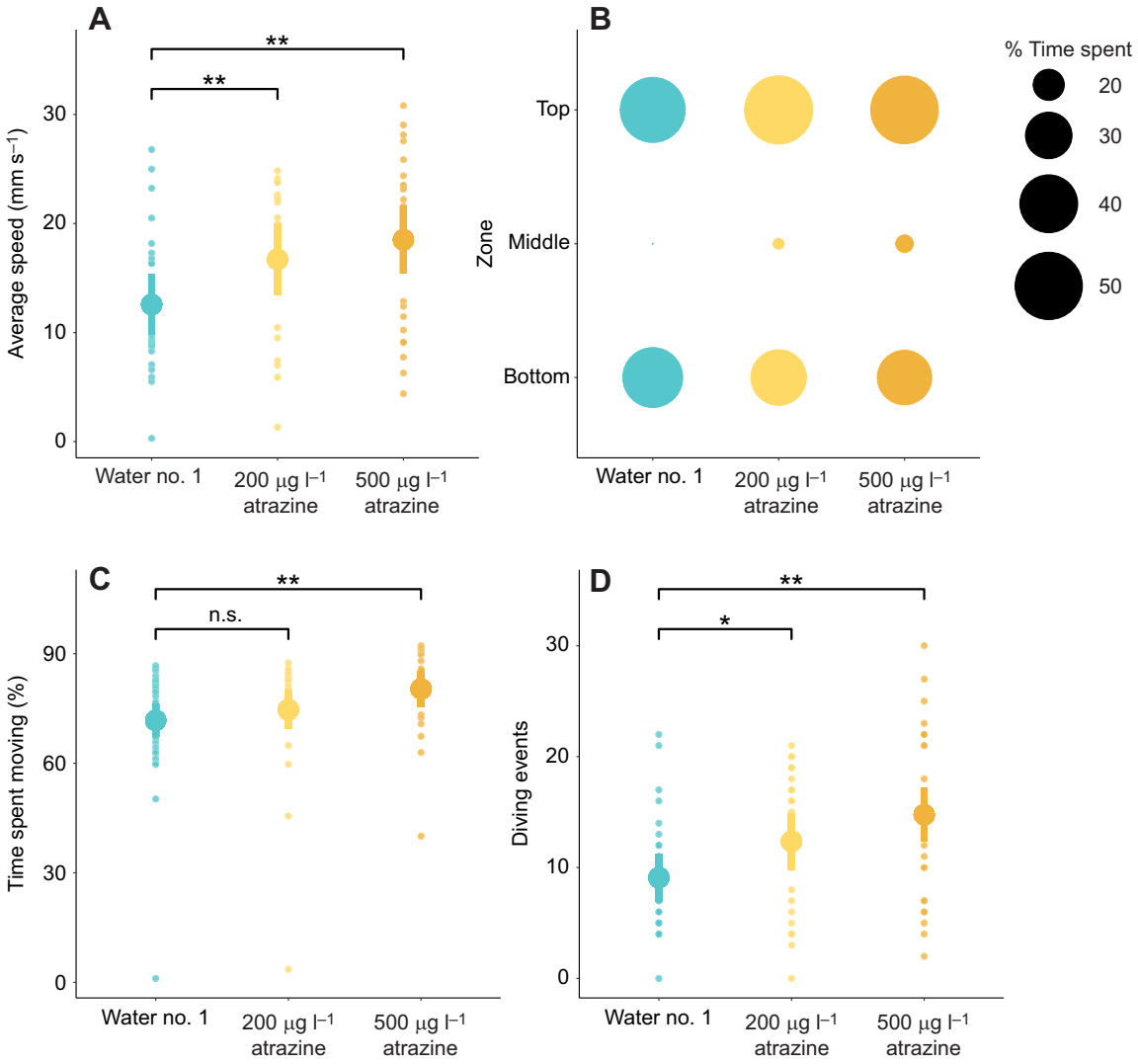


Fig. 3. Example of spontaneous locomotor activity for larvae reared in atrazine at field doses during familiarisation. (A) Average speed for each individual during the training period. (B) Average percentage time spent in each zone. (C) Average percentage time spent moving (TM). (D) Number of diving events. Large circles in A, C and D indicate mean values and bars indicate ±95% confidence intervals. Dark square indicates mean value for one treatment. n.s., not significant; *P<0.05, **P<0.01.

of animals provides an understanding of how pollutants may affect higher neurological functions as well as their environment (Bownik and Wlodkowic, 2021). This study investigated how three pollutants ranging from spray doses to realistic concentrations affect activity, learning and memory in mosquito larvae. Our results show that atrazine, glyphosate and paracetamol can alter spontaneous activity, impair habituation and disrupt memory retention as early as 2 h after exposure. Such changes may affect the larvae's ability to detect and respond appropriately to potential threats or to conserve energy by ignoring harmless stimuli in their natural habitat, ultimately reducing individual fitness.

Our findings provide evidence of behavioural changes caused by sub-lethal doses of pollutants in the water. This suggests that mosquito larvae may serve as effective bioindicators to study the effect of chemicals at sub-lethal doses. Moreover, we observed deleterious effects of glyphosate, atrazine and paracetamol presented in a mixture at doses at which no effect could be observed due to each substance alone. This means that our bioassay proved useful in revealing evidence of toxic effects of the mixture itself, also known as 'cocktail effects', which cannot be predicted solely by chemical analyses.

In this study, given that we were interested on sub-lethal hidden effects, we did not perform a survival analysis, as was previously done for glyphosate (Baglan et al., 2018). Although we cannot state that individual survival was not impacted at all, we did not observe any noticeable mortality during the rearing of the larvae. In the study by Baglan et al. (2018), the authors used glyphosate concentrations similar to ours, and no reduction in survival was found. In addition, a study by Bara et al. (2014) examined the effects of atrazine and glyphosate at 5 mg l^{-1} on life history traits in *Ae. aegypti* and *Aedes albopictus* mosquitoes: atrazine increased the emergence rate and the emergence time of *Ae. aegypti*, and also skewed the sex ratio. Another study (Rakotondravelo et al., 2006) assessed the sublethal effects of atrazine on survival, growth and adult emergence of the aquatic midge *Chironomus tentans*, and no effect on these three parameters at 150 µg l^{-1} was found. For the cockroach *Nauphoeta cinerea* (Adedara et al., 2021), no effect on individual survival was found when cockroaches had been administered 1.0 and 0.5 µg g^{-1} atrazine. Finally, Marcus and Fiumera (2016) found a reduction in survival in the fly *Drosophila melanogaster* starting at 2 ppb of atrazine dissolved in the diet. We did not find any previous

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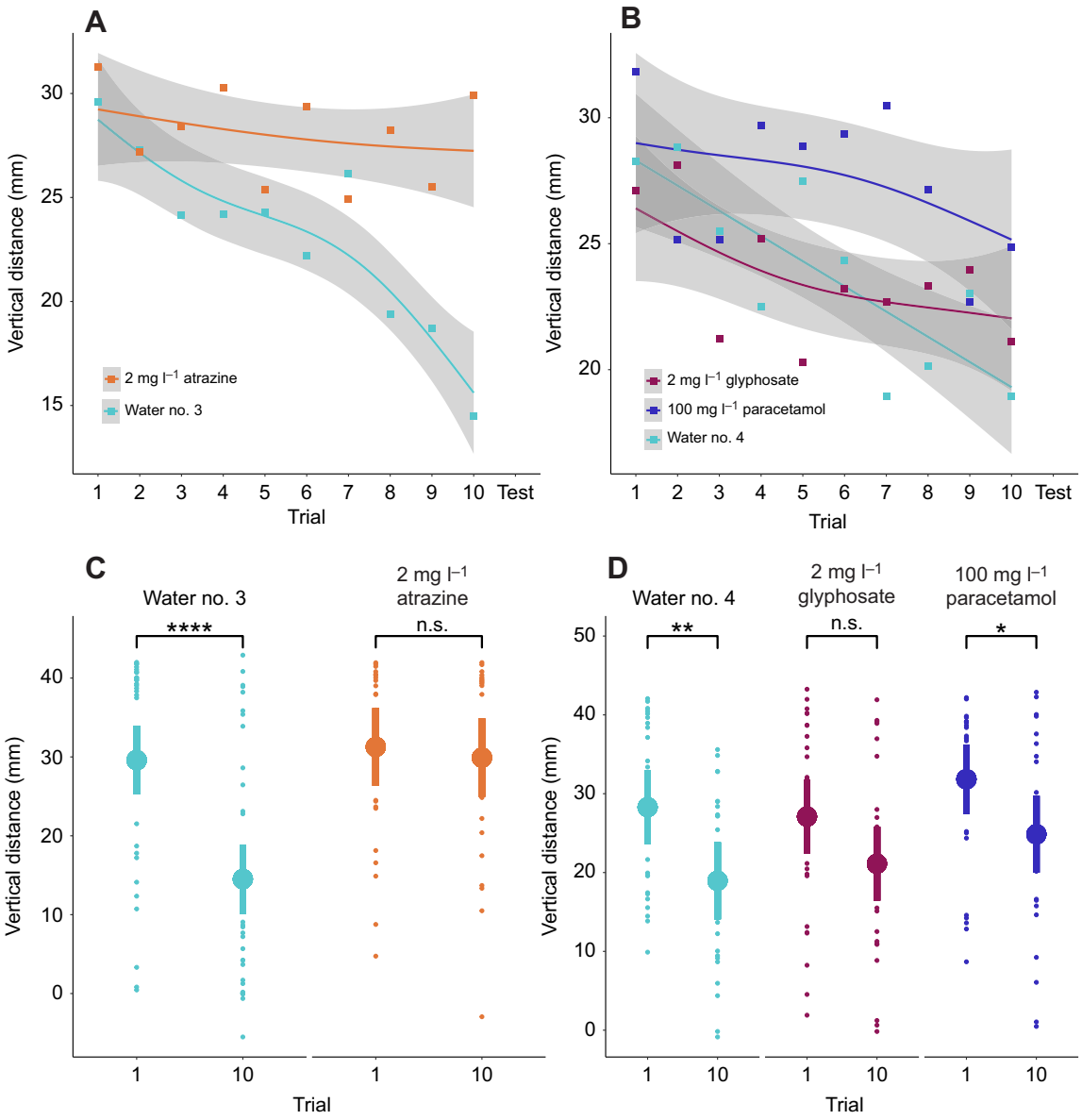


Fig. 4. Learning performance for larvae reared at spray doses. (A,B) Habituation curves for larvae reared in water treatment (cyan), 2 mg l^{-1} atrazine (orange; A), 2 mg l^{-1} glyphosate (red; B) and 100 mg l^{-1} paracetamol (blue; B). Vertical distance corresponds to the distance travelled by one individual during the stimulus period (3 s), from the 1st to the 10th trial. Smoothing lines indicate the best fitted GAM model. Squares indicate mean values and grey shading indicates the 95% confidence interval for the average response. (C,D) For each treatment, vertical distance travelled by individuals responding to an aversive stimulus during the 1st to the 10th trial. Large circles indicate mean values and bars indicate ±95% confidence intervals. n.s., not significant; **P*<0.05, ***P*<0.01, *****P*<0.0001.

studies on the effect of paracetamol on mosquito survival, and we did not observe any specific mortality in our study.

Atrazine affected our animals by altering their learning ability and by increasing their spontaneous locomotion at 500 µg l^{-1}. At 200 µg l^{-1}, atrazine slightly increased individual spontaneous activity but not at 2 mg l^{-1}. At the latter concentration, however, learning ability was severely impaired, as the average distance travelled during a stimulus was not less than 25 mm, whereas it fell below 20 mm in the water treatments. Furthermore, when looking at the response 2 h after the end of the training phase (i.e. acute toxicity), atrazine impaired the individual's ability to retrieve the information, whereas learning was not impaired. These three independent results provide strong evidence for the neurotoxicity of atrazine in mosquito larvae. As noted by several authors, atrazine disrupts endocrine functions in invertebrates, by increasing oxidative stress and the activity of enzymes such as P450 known to be involved in insecticide resistance in mosquitoes (Semren et al., 2018). Moreover, atrazine has been shown to alter acetylcholinesterase activity in invertebrate species (Boyer et al., 2006; Vogel et al., 2015).

Additionally, atrazine induces changes in locomotion, but with conflicting results from previous studies. For instance, it decreased locomotor activity in cockroaches, honeybees, nematodes and termites (García-Espiñeira, 2018; Ejomah et al., 2020; Araújo et al., 2021; Adedara et al., 2021), whereas in amphibians, it increased activity at low concentrations and decreased activity at high concentrations (Rohr and McCoy, 2010). In our study, we hypothesise that the hyperactivity induced at field concentrations was attenuated by greater toxicity at higher concentrations. To the

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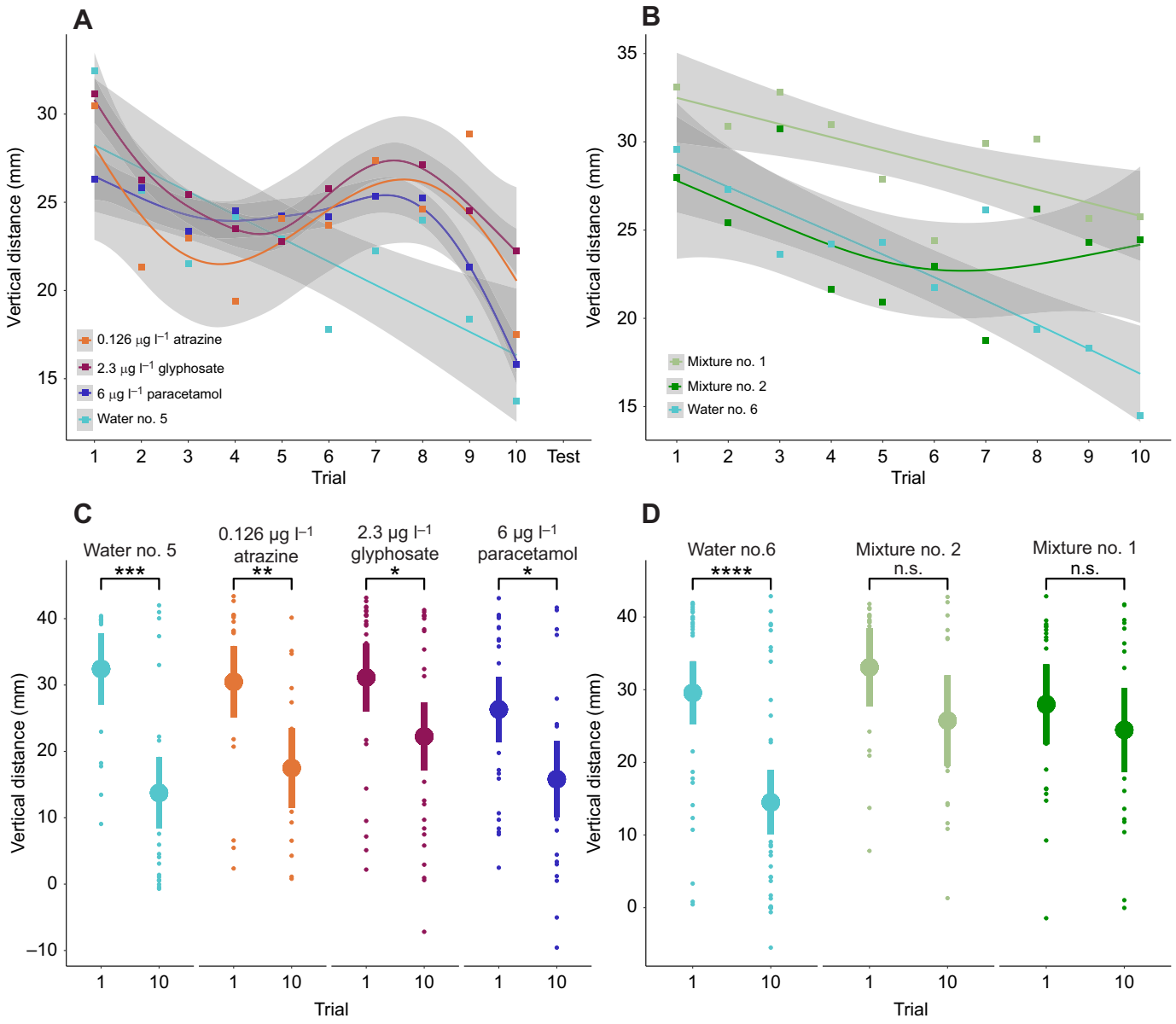


Fig. 5. Learning performance for larvae reared at field-realistic doses. (A,B) Habituation curves for larvae reared in water treatment (cyan), 0.126 µg l^{-1} atrazine (orange; A), 2.3 µg l^{-1} glyphosate (red; A), 6 µg l^{-1} paracetamol (blue; A), mixture no. 1 (200 µg l^{-1} atrazine+1 mg l^{-1} paracetamol; light green; B) and mixture no. 2 (0.126 µg l^{-1} atrazine+2.3 µg l^{-1} glyphosate+6 µg l^{-1} paracetamol; dark green; B). Vertical distance corresponds to the distance travelled by one individual during the stimulus period (3 s), from the 1st to the 10th trial. Smoothing lines indicate the best fitted GAM model. Squares indicate mean values and grey shading indicates the 95% confidence interval for the average response. (C,D) For each treatment, vertical distance travelled by individuals responding to an aversive stimulus during the 1st to the 10th trial. Large circles indicate mean values and bars indicate ±95% confidence intervals. n.s., not significant; *P<0.05, **P<0.01, ***P<0.001, ****P<0.0001.

best of our knowledge, the only studies of the effects of atrazine on learning and memory have been in mammals. For example, Rastegar-Moghaddam et al. (2019) found that atrazine ingestion impaired learning and spatial memory in mice and increased apoptosis of cells located in the hippocampus. Here, we showed for the first time that sub-lethal doses of atrazine impaired learning and memory in an invertebrate species.

Glyphosate also affected learning abilities of mosquito larvae even at the lowest dose: 100 µg l^{-1}. In addition, spray doses (5 mg l^{-1}) severely impaired individual learning and memory and increased locomotor activity. In acute toxicity tests, we also found a strong effect of glyphosate on memory 2 h after training. These results confirm the previously demonstrated effects of glyphosate on insect cognition and add to the growing body of work showing a negative effect of glyphosate on animals (Gill et al., 2018). Indeed, glyphosate reduced locomotor activity in nematodes, planarians and cockroaches and produced a slight and transient modification in bumblebees (García-Espiñeira, 2018; Córdova López et al., 2019; Kanabar et al., 2021; Nouvian et al., 2023). Glyphosate has also been shown to alter spatial learning in honeybees, associative learning in honeybees and bumblebees and memory retention in honeybees (Herbert et al., 2014; Balbuena et al., 2015; Hernández et al., 2021; Nouvian et al., 2023). We hypothesise, in line with the literature, that the alteration primarily affects the central nervous system of mosquito larvae (Gill et al., 2018; Baglan et al., 2018).

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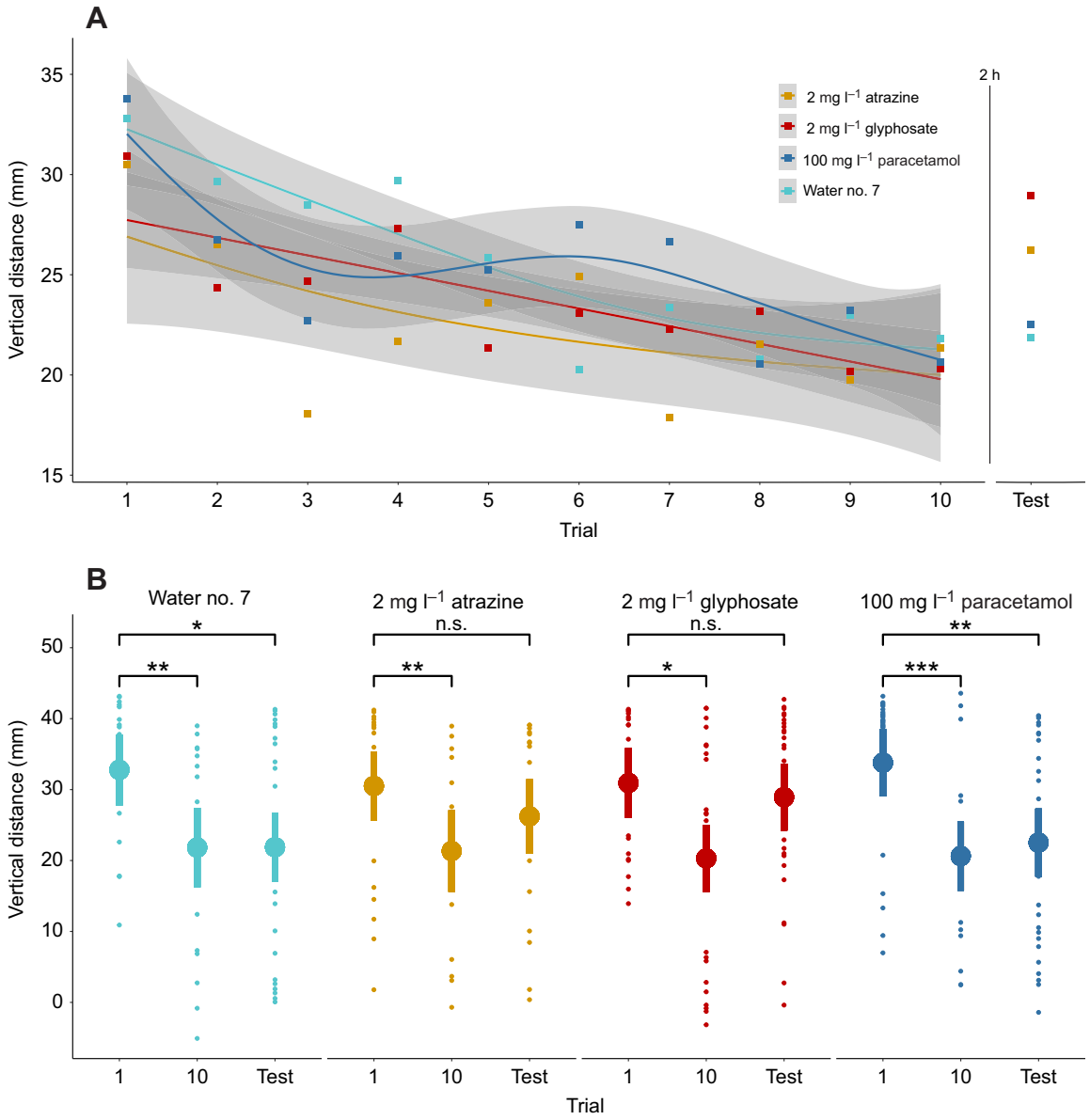


Fig. 6. Learning performance for larvae reared at spray doses for acute toxicity. (A) Habituation curves for larvae reared in water treatment (cyan), 2 mg l^{-1} atrazine (yellow), 2 mg l^{-1} glyphosate (red) and 100 mg l^{-1} paracetamol (blue). Vertical distance corresponds to the distance travelled by one individual during the stimulus period (3 s), from the 1st to the 10th trial and in an additional performed trial 2 h after the training period ('test'). Smoothing lines indicate the best fitted GAM model. Squares indicate mean values and grey shading indicates the 95% confidence interval for the average response. (B) For each treatment, vertical distance travelled by individuals responding to an aversive stimulus during the 1st to the 10th trial and in the additional test trial. Points indicate mean values and bars indicate ±95% confidence intervals. n.s., not significant; *P<0.05, **P<0.01, ***P<0.001.

Larvae reared at 10 mg l^{-1} paracetamol slightly decreased their behavioural response during the training period, but we observed a clear difference between the 1st and the 10th trials for larvae reared at 1 mg l^{-1} and 100 mg l^{-1}. In addition, no effect of paracetamol was found for acute toxicity, and only an increase in the number of diving events was observed at the spray doses for the activity assessment. The literature assessing the effects of paracetamol on locomotor and cognitive abilities in animals remains scarce. We found two studies in mice and rats that observed an alteration of learning abilities in the presence of paracetamol. Regarding locomotor activity, Viberg et al. (2014) reported mixed effects in locomotion in mice, whereas Nogueira et al. (2019) observed increased locomotion in zebrafish and Blecharz-Klin et al. (2017) found reduced motor activity in rats following paracetamol exposure. Our study cannot determine the exact effects of paracetamol on invertebrate cognition, but we suggest that research into the potential effects and mechanisms of this drug in the soil, water and organisms around human populations should continue (Hider–Mlynarz et al., 2018).

In our study, we combined the lowest concentrations of atrazine (200 µg l^{-1}) and paracetamol (1 mg l^{-1}). While these two concentrations alone had no effect, we found a significant increase in locomotion and an alteration in learning abilities in larvae reared with these two compounds in combination. Similarly, we conducted a series of experiments with the three compounds at field-realistic concentrations measured directly in water. Alone, the pollutants showed no effect on learning. However, when the pollutants were presented in a mixture, learning was impaired while no effect on locomotion was found. It is important to assess the

additive and possible synergistic effects of these compounds in mixtures. Indeed, aquatic organisms are constantly exposed to different types of pollutants and the possible synergistic effect of these pollutants would represent the worst case scenario for these organisms (Siviter et al., 2021). We chose to expose mosquito larvae to pure chemicals in order to assess the effect of the combination without the influence of adjuvants, but it is important to point out that commercial herbicide formulations affect organisms not only through their main active ingredient but also through their overall formulation and residual products (Córdova López et al., 2019; Kanabar et al., 2021).

Taken together, these results support the global concern about the lack of knowledge about the impact of agrochemicals combined with pharmaceutical residues on aquatic ecosystems. Mosquitoes are known for their behavioural plasticity and their tolerance to pesticides (Poupardin et al., 2008). Their wide global distribution, their tolerance of poor water conditions and their behaviour being affected by chemicals at low concentrations make them appropriate subjects to study the effects of pollutants on aquatic invertebrates. By altering the cognitive abilities of the larval stage, these pollutants increased energy expenditure by means of locomotor activity, which may affect their role in nutrient cycling or as food for predators, and affect the overall food web dynamic in aquatic ecosystems (Kanabar et al., 2021). As mosquitoes are resistant to stressors, these toxicological effects are of concern for other species living in the same environments and might be transferred by bioaccumulation to higher predators (Corbi et al., 2010).

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Competing interests
The authors declare no competing or financial interests.

Author contributions
Conceptualization: M.D., C.R.L., F.J.G.; Data curation: M.D.; Formal analysis: M.D.; Funding acquisition: F.J.G.; Investigation: M.D., C.R.L., F.J.G.; Methodology: M.D., C.R.L., F.J.G.; Project administration: M.D., C.R.L., F.J.G.; Resources: F.J.G.; Software: M.D.; Supervision: C.R.L., F.J.G.; Validation: M.D., C.R.L., F.J.G.; Visualization: M.D., C.R.L., F.J.G.; Writing – original draft: M.D., C.R.L., F.J.G.; Writing – review & editing: M.D., C.R.L., F.J.G.

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Data and resource availability
The R code used to analyse the data and the database are available from GitHub: https://github.com/martindessart/Chemical_toxicity. All other relevant data and details of resources can be found within the article and its supplementary information.

References
Abdulelah, S. A., Crile, K. G., Almouseli, A., Awali, S., Tutwiler, A. Y., Tien, E. A., Manzo, V. J., Hadeed, M. N. and Belanger, R. M. (2020). Environmentally relevant atrazine exposures cause DNA damage in cells of the lateral antennules of crayfish (*Faxonius virilis*). *Chemosphere* **239**, 124786. doi:10.1016/j.chemosphere.2019.124786

Adamson, G. M., Papadimitriou, J. M. and Harman, A. W. (1991). Postnatal mice have low susceptibility to paracetamol toxicity. *Pediatr. Res.* **29**, 496-499. doi:10.1203/00006450-199105010-00016

Adedara, I. A., Godswill, U.-A. S., Mike, M. A., Afolabi, B. A., Amorha, C. C., Sule, J., Rocha, J. B. T. and Farombi, E. O. (2021). Chronic ciprofloxacin and atrazine co-exposure aggravates locomotor and exploratory deficits in non-target detritivore speckled cockroach (*Nauphoeta cinerea*). *Environ. Sci. Pollut. Res.* **28**, 25680-25691. doi:10.1007/s11356-021-12460-5

Adegoke, A., Neff, E., Geary, A., Husser, M. C., Wilson, K., Norris, S. M., Dharmarajan, G. and Karim, S. (2020). Laboratory colonization by *Dirofilaria immitis* alters the microbiome of female Aedes aegypti mosquitoes. *Parasites Vectors* **13**, 349. doi:10.1186/s13071-020-04218-8

Araújo, R. S., Bernardes, R. C. and Martins, G. F. (2021). A mixture containing the herbicides Mesotrione and Atrazine imposes toxicological risks on workers of *Partamona helleri*. *Sci. Total Environ.* **763**, 142980. doi:10.1016/j.scitotenv.2020.142980

Arthington, A. H. (2021). Grand challenges to support the freshwater biodiversity emergency recovery plan. *Front. Environ. Sci.* **9**, 664313. doi:10.3389/fenvs.2021.664313

Baglan, H., Lazzari, C. and Guerrieri, F. (2017). Learning in mosquito larvae (*Aedes aegypti*): Habituation to a visual danger signal. *J. Insect Physiol.* **98**, 160-166. doi:10.1016/j.jinsphys.2017.01.001

Baglan, H., Lazzari, C. R. and Guerrieri, F. J. (2018). Glyphosate impairs learning in *Aedes aegypti* mosquito larvae at field-realistic doses. *J. Exp. Biol.* **221**, jeb187518. doi:10.1242/jeb.187518

Balbuena, M. S., Tison, L., Hahn, M.-L., Greggers, U., Menzel, R. and Farina, W. M. (2015). Effects of sublethal doses of glyphosate on honeybee navigation. *J. Exp. Biol.* **218**, 2799-2805. doi:10.1242/jeb.117291

Bara, J. J., Montgomery, A. and Muturi, E. J. (2014). Sublethal effects of atrazine and glyphosate on life history traits of *Aedes aegypti* and *Aedes albopictus* (Diptera: Culicidae). *Parasitol. Res.* **113**, 2879-2886. doi:10.1007/s00436-014-3949-y

Bartram, J. and Ballance, R. (eds) (1996). *Water Quality Monitoring: A Practical Guide to the Design and Implementation of Freshwater Quality studies and Monitoring Programmes*, 1st edn. London: CRC Press.

Bataillard, D., Christe, P. and Pigeault, R. (2020). Impact of field-realistic doses of glyphosate and nutritional stress on mosquito life history traits and susceptibility to malaria parasite infection. *Ecol Evol* **10**, 5079-5088. doi:10.1002/ece3.6261

Battaglin, W. A., Meyer, M. T., Kuivila, K. M. and Dietze, J. E. (2014). Glyphosate and its degradation product AMPA occur frequently and widely in U.S. soils, surface water, groundwater, and precipitation. *J. American Water Resour. Assoc.* **50**, 275-290. doi:10.1111/jawr.12159

Bertram, M. G., Martin, J. M., McCallum, E. S., Alton, L. A., Brand, J. A., Brooks, B. W., Cerveny, D., Fick, J., Ford, A. T., Hellström, G. et al. (2022). Frontiers in quantifying wildlife behavioural responses to chemical pollution. *Biol. Rev.* **97**, 1346-1364. doi:10.1111/brv.12844

Black, W. C., Snell, T. K., Saavedra-Rodriguez, K., Kading, R. C. and Campbell, C. L. (2021). From global to local—new insights into features of pyrethroid detoxification in vector mosquitoes. *Insects* **12**, 276. doi:10.3390/insects12040276

Blaxter, J. H. S. and Hallers-Tjabbes, C. C. T. (1992). The effect of pollutants on sensory systems and behaviour of aquatic animals. *Netherlands Journal of Aquatic Ecology* **26**, 43-58. doi:10.1007/BF02298027

Blecharz-Klin, K., Piechal, A., Jawna-Zboińska, K., Pyrzanowska, J., Wawer, A., Joniec-Maciejak, I. and Widy-Tyszkiewicz, E. (2017). Paracetamol–Effect of early exposure on neurotransmission, spatial memory and motor performance in rats. *Behav. Brain Res.* **323**, 162-171. doi:10.1016/j.bbr.2017.01.051

Bonada, N., Prat, N., Resh, V. H. and Statzner, B. (2006). Developments in aquatic insect biomonitoring: a comparative analysis of recent approaches. *Annu. Rev. Entomol.* **51**, 495-523. doi:10.1146/annurev.ento.51.110104.151124

Bownik, A. and Wlodkowic, D. (2021). Applications of advanced neuro-behavioral analysis strategies in aquatic ecotoxicology. *Sci. Total Environ.* **772**, 145577. doi:10.1016/j.scitotenv.2021.145577

Boyer, S., Sérandour, J., Lempérière, G., Raveton, M. and Ravanel, P. (2006). Do herbicide treatments reduce the sensitivity of mosquito larvae to insecticides? *Chemosphere* **65**, 721-724. doi:10.1016/j.chemosphere.2006.02.032

Byer, J. D., Struger, J., Klawunn, P., Todd, A. and Sverko, E. (2008). Low cost monitoring of glyphosate in surface waters using the ELISA method: an evaluation. *Environ. Sci. Technol.* **42**, 6052-6057. doi:10.1021/es8005207

Corbi, J. J., Froehlich, C. G., Strixino, S. T. and Santos, A. D. (2010). Bioaccumulation of metals in aquatic insects of streams located in areas with sugar cane cultivation. *Quím. Nova* **33**, 644-648. doi:10.1590/S0100-40422010000300030

Córdova López, A. M., Sarmento, R. A., de Souza Saraiva, A., Pereira, R. R., Soares, A. M. V. M. and Pestana, J. L. T. (2019). Exposure to Roundup® affects behaviour, head regeneration and reproduction of the freshwater planarian *Girardia tigrina*. *Sci. Total Environ.* **675**, 453-461. doi:10.1016/j.scitotenv.2019.04.234

Dessart, M., Piñeirúa, M., Lazzari, C. R. and Guerrieri, F. J. (2023). Assessing learning in mosquito larvae using video-tracking. *J. Insect Physiol.* **149**, 104535. doi:10.1016/j.jinsphys.2023.104535

Dessart, M., Lazzari, C. R. and Guerrieri, F. J. (2024). Habituation leads to short but not long term memory formation in mosquito larvae. *J. Insect Physiol.* **155**, 104650. doi:10.1016/j.jinsphys.2024.104650

Dewey, S. L. (1986). Effects of the herbicide atrazine on aquatic insect community structure and emergence. *Ecology* **67**, 148-162. doi:10.2307/1938513

Duong, M., Gulmez, S. E., Salvo, F., Abouelfath, A., Lassalle, R., Droz, C., Blin, P. and Moore, N. (2016). Usage patterns of paracetamol in France. *Br. J. Clin. Pharmacol.* **82**, 498-503. doi:10.1111/bcp.12957

Ejomah, A. J., Uyi, O. O. and Ekaye, S.-O. (2020). Exposure of the African mound building termite, *Macrotermes bellicosus* workers to commercially formulated 2,4-D and atrazine caused high mortality and impaired locomotor response. *PLoS ONE* **15**, e0230664. doi:10.1371/journal.pone.0230664

El Menyiy, N., Al-Waili, N., El Ghouizi, A., Al-Waili, W. and Lyoussi, B. (2018). Evaluation of antiproteinuric and hepato-renal protective activities of propolis in paracetamol toxicity in rats. *Nutr. Res. Pract.* **12**, 535. doi:10.4162/nrp.2018.12.6.535

Environmental Protection Agency (1994). Using toxicity tests in ecological risk assessment, Vol. 2, No.1 Publication 9345.0-05I. https://www.epa.gov/risk/using-toxicity-tests-ecological-risk-assessment.

Ford, A. T., Ågerstrand, M., Brooks, B. W., Allen, J., Bertram, M. G., Brodin, T., Dang, Z., Duquesne, S., Sahm, R., Hoffmann, F. et al. (2021). The role of behavioral ecotoxicology in environmental protection. *Environ. Sci. Technol.* **55**, 5620-5628. doi:10.1021/acs.est.0c06493

García-Espiñeira, M. (2018). Toxicity of atrazine- and glyphosate-based formulations on *Caenorhabditis elegans*. *Ecotoxicol. Environ. Saf.* **156**, 216-222. doi:10.1016/j.ecoenv.2018.02.075

Gill, J. P. K., Sethi, N., Mohan, A., Datta, S. and Girdhar, M. (2018). Glyphosate toxicity for animals. *Environ. Chem. Lett.***16**, 401-426. doi:10.1007/s10311-017-0689-0

Gilliom, R. J., Barbash, J. E., Crawford, C. G., Hamilton, P. A., Martin, J. D., Nakagaki, N., Nowell, L. H., Scott, J. C., Stackelberg, P. E., Thelin, G. P. and Wolock, D. M. (2006). Pesticides in the nation's streams and ground water, 1992–2001. U.S. Geological Survey Circular 1291, The Quality of Our Nation's Waters, 184 pp. U.S. Geological Survey. https://pubs.er.usgs.gov/publication/cir1291

Gracia-Lor, E., Sancho, J. V., Serrano, R. and Hernández, F. (2012). Occurrence and removal of pharmaceuticals in wastewater treatment plants at the Spanish Mediterranean area of Valencia. *Chemosphere* **87**, 453-462. doi:10.1016/j.chemosphere.2011.12.025

Hellou, J. (2011). Behavioural ecotoxicology, an "early warning" signal to assess environmental quality. *Environ. Sci. Pollut. Res.* **18**, 1-11. doi:10.1007/s11356-010-0367-2

Herbert, L. T., Vazquez, D. E., Arenas, A. and Farina, W. M. (2014). Effects of field-realistic doses of glyphosate on honeybee appetitive behaviour. *J. Exp. Biol.* **217**, 3457-3464.

Hernández, J., Riveros, A. J. and Amaya-Márquez, M. (2021). Sublethal doses of glyphosate impair olfactory memory retention, but not learning in the honey bee (*Apis mellifera scutellata*). *J. Insect Conserv.* **25**, 683-694. doi:10.1007/s10841-021-00335-6

Hider-Mlynarz, K., Cavalié, P. and Maison, P. (2018). Trends in analgesic consumption in France over the last 10 years and comparison of patterns across Europe. *Br. J. Clin. Pharmacol.* **84**, 1324-1334. doi:10.1111/bcp.13564

Hodkinson, I. D. and Jackson, J. K. (2005). Terrestrial and aquatic invertebrates as bioindicators for environmental monitoring, with particular reference to mountain ecosystems. *Environ. Manag.* **35**, 649-666. doi:10.1007/s00267-004-0211-x

Hong, X. and Zha, J. (2019). Fish behavior: a promising model for aquatic toxicology research. *Sci. Total Environ.* **686**, 311-321. doi:10.1016/j.scitotenv.2019.06.028

Jacquin, L., Petitjean, Q., Côte, J., Laffaille, P. and Jean, S. (2020). Effects of pollution on fish behavior, personality, and cognition: some research perspectives. *Front. Ecol. Evol.* **8**, 86. doi:10.3389/fevo.2020.00086

Johnson, J. A., Ihunwo, A., Chimuka, L. and Mbajiorgu, E. F. (2019). Cardiotoxicity in African clawed frog (*Xenopus laevis*) sub-chronically exposed to environmentally relevant atrazine concentrations. Implications for species survival. *Aquat. Toxicol.* **213**, 105218. doi:10.1016/j.aquatox.2019.06.001

Kanabar, M., Bauer, S., Ezedum, Z. M., Dwyer, I. P., Moore, W. S., Rodriguez, G., Mall, A., Littleton, A. T., Yudell, M., Kanabar, J. et al. (2021). Roundup negatively impacts the behavior and nerve function of the Madagascar hissing cockroach (*Gromphadorhina portentosa*). *Environ. Sci. Pollut. Res.* **28**, 32933-32944. doi:10.1007/s11356-021-13021-6

Kraemer, M. U. G., Reiner, R. C., Brady, O. J., Messina, J. P., Gilbert, M., Pigott, D. M., Yi, D., Johnson, K., Earl, L., Marczak, L. B. et al. (2019). Past and future spread of the arbovirus vectors *Aedes aegypti* and *Aedes albopictus*. *Nat. Microbiol.* **4**, 854-863. doi:10.1038/s41564-019-0376-y

Ledieu, L., Simonneau, A., Thiebault, T., Fougere, L., Destandau, E., Cerdan, O. and Laggoun, F. (2021). Spatial distribution of pharmaceuticals within the particulate phases of a peri-urban stream. *Chemosphere* **279**, 130385. doi:10.1016/j.chemosphere.2021.130385

Li, J., Li, X., Bi, H., Ma, K. and Li, B. (2018). Developmental exposure to atrazine impairs spatial memory and downregulates the hippocampal D1 dopamine receptor and cAMP-dependent signaling pathway in rats. *IJMS* **19**, 2241. doi:10.3390/ijms19082241

Li, X., Kong, H., Ji, X., Gao, Y. and Jin, M. (2019). Zebrafish behavioral phenomics applied for phenotyping aquatic neurotoxicity induced by lead contaminants of environmentally relevant level. *Chemosphere* **224**, 445-454. doi:10.1016/j.chemosphere.2019.02.174

Lima, I. B., Boëchat, I. G., Fernandes, M. D., Monteiro, J. A. F., Rivaroli, L. and Gücker, B. (2022). Glyphosate pollution of surface runoff, stream water, and drinking water resources in Southeast Brazil. *Environ. Sci. Pollut. Res.* **30**, 27030-27040. doi:10.1007/s11356-022-24167-2

López-López, E. and Sedeño-Díaz, J. E. (2015). Biological indicators of water quality: the role of fish and macroinvertebrates as indicators of water quality. In *Environmental Indicators* (ed. R. H. Armon and O. Hänninen), pp. 643-661. Dordrecht: Springer Netherlands.

Madhav, S., Ahamad, A., Singh, A. K., Kushawaha, J., Chauhan, J. S., Sharma, S. and Singh, P. (2020). Water pollutants: sources and impact on the environment and human health. *In Sensors in Water Pollutants Monitoring: Role of Material* (ed. D. Pooja, P. Kumar, P. Singh and and S. Patil), pp. 43-62. Singapore: Springer Singapore.

Malafaia, G. and Da Luz, T. M. (2024). Behavioural endpoints of Nile Tilapia (*Oreochromis niloticus*): potential biomarkers for evaluating the ecotoxicity of contaminants of emerging concern and legacy pollutants. In *Fish Species in Environmental Risk Assessment Strategies* (ed. G. E. Liwszyc and M. L. Larramendy), pp. 162-202. Royal Society of Chemistry.

Malhotra, N., Chen, K. H.-C., Huang, J.-C., Lai, H.-T., Uapipatanakul, B., Roldan, M. J. M., Macabeo, A. P. G., Ger, T.-R. and Hsiao, C.-D. (2021). Physiological effects of neonicotinoid insecticides on non-target aquatic animals—an updated review. *IJMS* **22**, 9591. doi:10.3390/ijms22179591

Marcus, S. R. and Fiumera, A. C. (2016). Atrazine exposure affects longevity, development time and body size in *Drosophila melanogaster*. *J. Insect Physiol.* **91-92**, 18-25. doi:10.1016/j.jinsphys.2016.06.006

McCrae, J. C., Morrison, E. E., MacIntyre, I. M., Dear, J. W. and Webb, D. J. (2018). Long-term adverse effects of paracetamol – a review. *Br. J. Clin. Pharmacol.* **84**, 2218-2230. doi:10.1111/bcp.13656

Melvin, S. D. and Wilson, S. P. (2013). The utility of behavioral studies for aquatic toxicology testing: a meta-analysis. *Chemosphere* **93**, 2217-2223. doi:10.1016/j.chemosphere.2013.07.036

Nagy, K., Duca, R. C., Lovas, S., Creta, M., Scheepers, P. T. J., Godderis, L. and Ádám, B. (2020). Systematic review of comparative studies assessing the toxicity of pesticide active ingredients and their product formulations. *Environ. Res.* **181**, 108926. doi:10.1016/j.envres.2019.108926

Nogueira, A. F., Pinto, G., Correia, B. and Nunes, B. (2019). Embryonic development, locomotor behavior, biochemical, and epigenetic effects of the pharmaceutical drugs paracetamol and ciprofloxacin in larvae and embryos of *Danio rerio* when exposed to environmental realistic levels of both drugs. *Environ. Toxicol.* **34**, 1177-1190. doi:10.1002/tox.22819

Nouvian, M., Foster, J. J. and Weidenmüller, A. (2023). Glyphosate impairs aversive learning in bumblebees. *Sci. Total Environ.* **898**, 165527. doi:10.1016/j.scitotenv.2023.165527

O'Brien, C. and Turner, B. (2004). Impact of paracetamol on *Calliphora vicina* larval development. *Int. J. Legal Med.* **118**, 188-189. doi:10.1007/s00414-004-0440-9

Poupardin, R., Reynaud, S., Strode, C., Ranson, H., Vontas, J. and David, J.-P. (2008). Cross-induction of detoxification genes by environmental xenobiotics and insecticides in the mosquito *Aedes aegypti*: Impact on larval tolerance to chemical insecticides. *Insect Biochem. Mol. Biol.* **38**, 540-551. doi:10.1016/j.ibmb.2008.01.004

Previšić, A., Rožman, M., Mor, J.-R., Acuña, V., Serra-Compte, A., Petrović, M. and Sabater, S. (2020). Aquatic macroinvertebrates under stress: bioaccumulation of emerging contaminants and metabolomics implications. *Sci. Total Environ.* **704**, 135333. doi:10.1016/j.scitotenv.2019.135333

Rakotondravelo, M. L., Anderson, T. D., Charlton, R. E. and Zhu, K. Y. (2006). Sublethal effects of three pesticides on larval survivorship, growth, and macromolecule production in the aquatic midge, *Chironomus tentans* (Diptera: Chironomidae). *Arch. Environ. Contam. Toxicol.* **51**, 352-359. doi:10.1007/s00244-005-0219-0

Rastegar-Moghaddam, S. H., Mohammadipour, A., Hosseini, M., Bargi, R. and Ebrahimzadeh-Bideskan, A. (2019). Maternal exposure to atrazine induces the hippocampal cell apoptosis in mice offspring and impairs their learning and spatial memory. *Toxin Rev.* **38**, 298-306. doi:10.1080/15569543.2018.1466804

Riaz, M. A., Poupardin, R., Reynaud, S., Strode, C., Ranson, H. and David, J.-P. (2009). Impact of glyphosate and benzo[a]pyrene on the tolerance of mosquito larvae to chemical insecticides. Role of detoxification genes in response to xenobiotics. *Aquat. Toxicol.* **93**, 61-69. doi:10.1016/j.aquatox.2009.03.005

Rohr, J. R. and McCoy, K. A. (2010). A qualitative meta-analysis reveals consistent effects of atrazine on freshwater fish and amphibians. *Environ. Health Perspect.* **118**, 20-32. doi:10.1289/ehp.0901164

Rubio-Palis, Y., Dzuris, N., Sandi, C., Vizcaino-Cabarrus, R. L., Corredor-Medina, C., González, J. A. and Lenhart, A. E. (2023). Insecticide resistance levels and associated mechanisms in three *Aedes aegypti* populations from Venezuela. *Mem. Inst. Oswaldo Cruz* **118**, e220210. doi:10.1590/0074-02760220210

Sala, O. E., Stuart Chapin, F., , 3rd, Armesto, J. J., Berlow, E., Bloomfield, J., Dirzo, R., Huber-Sanwald, E., Huenneke, L. F., Jackson, R. B., Jackson, R. B. et al. (2000). Global biodiversity scenarios for the year 2100. *Science* **287**, 1770-1774. doi:10.1126/science.287.5459.1770

Semren, T. Ž., Žunec, S. and Pizent, A. (2018). Oxidative stress in triazine pesticide toxicity: a review of the main biomarker findings. *Arch. Ind. Hygiene Toxicol.* **69**, 109-125. doi:10.2478/aiht-2018-69-3118

Siviter, H., Bailes, E. J., Martin, C. D., Oliver, T. R., Koricheva, J., Leadbeater, E. and Brown, M. J. F. (2021). Agrochemicals interact synergistically to increase bee mortality. *Nature* **596**, 389-392. doi:10.1038/s41586-021-03787-7

Struger, J., Thompson, D., Staznik, B., Martin, P., McDaniel, T. and Marvin, C. (2008). Occurrence of glyphosate in surface waters of southern Ontario. *Bull. Environ. Contam. Toxicol.* **80**, 378-384. doi:10.1007/s00128-008-9373-1

Upshur, I. F., Fehlman, M., Parikh, V., Vinauger, C. and Lahondère, C. (2023). Sugar feeding by invasive mosquito species on ornamental and wild plants. *Sci. Rep.* **13**, 22121. doi:10.1038/s41598-023-48089-2

Vári, Á., Podschun, S. A., Erős, T., Hein, T., Pataki, B., Iojă, I.-C., Adamescu, C. M., Gerhardt, A., Gruber, T., Dedić, A. et al. (2022). Freshwater systems and ecosystem services: Challenges and chances for cross-fertilization of disciplines. *Ambio* **51**, 135-151. doi:10.1007/s13280-021-01556-4

Viberg, H., Eriksson, P., Gordh, T. and Fredriksson, A. (2014). Paracetamol (Acetaminophen) administration during neonatal brain development affects cognitive function and alters its analgesic and anxiolytic response in adult male mice. *Toxicol. Sci.* **138**, 139-147. doi:10.1093/toxsci/kft329

Vogel, A., Jocque, H., Sirot, L. K. and Fiumera, A. C. (2015). Effects of atrazine exposure on male reproductive performance in *Drosophila melanogaster*. *J. Insect Physiol.* **72**, 14-21. doi:10.1016/j.jinsphys.2014.11.002

Weidenmüller, A., Meltzer, A., Neupert, S., Schwarz, A. and Kleineidam, C. (2022). Glyphosate impairs collective thermoregulation in bumblebees. *Science* **376**, 1122-1126. doi:10.1126/science.abf7482

Wood, S. N. (2017). *Generalized Additive Models: An Introduction With R*, 2nd edn. Chapman and Hall/CRC.

Zaghloul, A., Saber, M., Gadow, S. and Awad, F. (2020). Biological indicators for pollution detection in terrestrial and aquatic ecosystems. *Bull. Natl. Res. Cent.* **44**, 127. doi:10.1186/s42269-020-00385-x

Zhang, X.-T., Chen, R.-S., Liu, G.-H., Yang, Y. and Feng, T.-W. (2021). Economic value of freshwater provisioning services of the cryosphere in the Urumqi River, Northwest China. *Adv. Clim. Chang. Res.* **12**, 894-902. doi:10.1016/j.accre.2021.09.003

Zhou, Q., Zhang, J., Fu, J., Shi, J. and Jiang, G. (2008). Biomonitoring: An appealing tool for assessment of metal pollution in the aquatic ecosystem. *Anal. Chim. Acta* **606**, 135-150. doi:10.1016/j.aca.2007.11.018

Summary: Examination of the effects of glyphosate, atrazine and paracetamol (acetaminophen) at field-realistic to commercial concentrations on mosquito larvae cognitive ability, introducing a novel approach to assess ecological risks in vulnerable ecosystems.

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